

COMPANION FOR CHAPTER 8

OF

**FUNDAMENTALS
OF PROBABILITY**

WITH STOCHASTIC PROCESSES

FOURTH EDITION

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Chapter 8

Bivariate Distributions

8A ADDITIONAL EXAMPLES

Example 8a The joint probability density function of the continuous random variables X and Y is given by

$$f(x, y) = \begin{cases} 12x(1 - x) & 0 \leq x \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X | Y = y)$.

Solution: Note that

$$E(X | Y = y) = \int_0^y x f_{X|Y}(x|y) dx,$$

where,

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}.$$

Now

$$f_Y(y) = \int_0^y 12x(1 - x) dx = -4y^3 + 6y^2, \quad 0 \leq y \leq 1.$$

Thus

$$f_{X|Y}(x|y) = \frac{12x(1 - x)}{-4y^3 + 6y^2}, \quad 0 \leq y \leq 1, 0 \leq x \leq y.$$

So

$$E(X | Y = y) = \int_0^y \frac{12x^2(1 - x)}{-4y^3 + 6y^2} dx = \frac{-3y^2 + 4y}{-4y + 6}, \quad 0 \leq y \leq 1. \quad \blacklozenge$$

Example 8b Let X be the fraction of lead and Y be the fraction of zinc in a random piece of a certain sedimentary rock. Suppose that the joint probability density function of

X and Y is given by

$$f(x, y) = \begin{cases} \frac{16}{3}(7x + 2y) & x > 0, y > 0, 2x + 2y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X)$ and $E(Y)$.

Solution: We have that

$$\begin{aligned} E(X) &= \frac{16}{3} \int_0^{1/2} \left[\int_0^{(1-2x)/2} x(7x + 2y) dy \right] dx \\ &= \frac{4}{3} \int_0^{1/2} x(1 - 2x)(12x + 1) dx = \frac{2}{9} \approx 0.22. \end{aligned}$$

To find $E(Y)$, similarly to the above calculation, we can use the formula

$$E(Y) = \frac{16}{3} \int_0^{1/2} \left[\int_0^{(1-2y)/2} y(7x + 2y) dx \right] dy.$$

However, for calculational simplicity, it is better to proceed as follows

$$\begin{aligned} E(Y) &= \frac{16}{3} \int_0^{1/2} \left[\int_0^{(1-2y)/2} y(7x + 2y) dx \right] dy \\ &= \frac{4}{3} \int_0^{1/2} y(1 - 2y)(7 - 10y) dy = \frac{1}{4} \approx 0.25. \quad \blacklozenge \end{aligned}$$

Example 8c A point is selected at random from the unit disk

$$D = \{(x, y) : x^2 + y^2 \leq 1\}.$$

Find the expected value of the distance of the point selected from the origin.

Solution: Let X and Y be the coordinates of the point selected. Let $f(x, y)$ be the joint probability density function of X and Y . By the discussion right before Definition 8.5 of the book,

$$f(x, y) = \begin{cases} \frac{1}{\text{area}(D)} & \text{if } (x, y) \in D \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$f(x, y) = \begin{cases} \frac{1}{\pi} & \text{if } x^2 + y^2 \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

The distance of the point selected from the origin is $\sqrt{x^2 + y^2}$. So we are interested in $E(\sqrt{x^2 + y^2})$, which is calculated as follows.

$$E(\sqrt{x^2 + y^2}) = \iint_{x^2 + y^2 \leq 1} \sqrt{x^2 + y^2} \cdot \frac{1}{\pi} dx dy.$$

To calculate this double integral, we will convert it to polar coordinates. We have $x = r \cos \theta$, $y = r \sin \theta$, $\sqrt{x^2 + y^2} = r$, $dx dy = r dr d\theta$, $0 \leq r \leq 1$, $0 \leq \theta < 2\pi$. So

$$\begin{aligned} E(\sqrt{x^2 + y^2}) &= E(r) = \int_0^{2\pi} \left[\int_0^1 r \cdot \frac{1}{\pi} r dr \right] d\theta \\ &= \frac{1}{\pi} \int_0^{2\pi} \left[\int_0^1 r^2 dr \right] d\theta = \frac{1}{\pi} \int_0^{2\pi} \frac{1}{3} d\theta = \frac{2}{3}. \quad \blacklozenge \end{aligned}$$

Example 8d An engineer, monitoring a system, identifies when a fault occurs and determines whether the fault is Type I or another type. Suppose that Y , the total number of faults occurring in an hour is a Poisson random variable with mean 3. Suppose that, independently of other faults, the probability that a fault is Type I is $1/3$. Let X be the number of Type I faults in an hour. Find $p(i, j)$, the joint probability mass function of X and Y , and $p_X(i)$, the marginal probability mass function of X .

Solution: By the law of multiplication,

$$\begin{aligned} p(i, j) &= P(X = i, Y = j) = P(X = i | Y = j)P(Y = j) \\ &= \binom{j}{i} \left(\frac{1}{3}\right)^i \left(\frac{2}{3}\right)^{j-i} \cdot \frac{e^{-3} \cdot 3^j}{j!} \\ &= \frac{e^{-3} \cdot 2^{j-i}}{i! (j-i)!}, \quad j = 0, 1, 2, \dots; 0 \leq i \leq j. \end{aligned}$$

The marginal probability mass function of X is

$$\begin{aligned} p_X(i) &= \sum_{j=i}^{\infty} p(i, j) = \sum_{j=i}^{\infty} \frac{e^{-3} \cdot 2^{j-i}}{i! (j-i)!} \\ &= \frac{e^{-3}}{i!} \sum_{j=i}^{\infty} \frac{2^{j-i}}{(j-i)!} = \frac{e^{-3}}{i!} \sum_{k=0}^{\infty} \frac{2^k}{k!} = \frac{e^{-3}}{i!} \cdot e^2 = \frac{e^{-1}}{i!}, \quad i = 0, 1, 2, \dots \end{aligned}$$

Therefore, the marginal probability mass function of X is Poisson with parameter 1. $\blacklozenge$

Example 8e Suppose that in a sequence of random digits the total number of 0's among the first 100 is 12. What is the expected number of 0's among the first 10?

Solution: Let X and Y denote the number of 0's among the first 10 and the first 100 digits, respectively. To find the desired quantity, $E(X | Y = 12)$, note that

$$E(X | Y = 12) = \sum_{x=0}^{10} x p_{X|Y}(x|12) = \sum_{x=0}^{10} x \frac{p(x, 12)}{p_Y(12)},$$

where

$$\begin{aligned} p(x, 12) &= P(X = x, Y = 12) = P(X = x, Y - X = 12 - x) \\ &= P(X = x)P(Y - X = 12 - x) \\ &= \binom{10}{x} \left(\frac{1}{10}\right)^x \left(\frac{9}{10}\right)^{10-x} \cdot \binom{90}{12-x} \left(\frac{1}{10}\right)^{12-x} \left(\frac{9}{10}\right)^{90-(12-x)} \\ &= \binom{10}{x} \binom{90}{12-x} \left(\frac{1}{10}\right)^{12} \left(\frac{9}{10}\right)^{88}. \end{aligned}$$

This is because X and $Y - X$, being the number of 0's among the first 10 and the number of 0's among the next 90 digits, are independent binomial random variables with the parameters $(10, 1/10)$ and $(90, 1/10)$, respectively. Therefore,

$$\begin{aligned} p_{X|Y}(x|12) &= \frac{p(x, 12)}{p_Y(12)} = \frac{\binom{10}{x} \binom{90}{12-x} \left(\frac{1}{10}\right)^{12} \left(\frac{9}{10}\right)^{88}}{\binom{100}{12} \left(\frac{1}{10}\right)^{12} \left(\frac{9}{10}\right)^{88}} \\ &= \frac{\binom{10}{x} \binom{90}{12-x}}{\binom{100}{12}}. \end{aligned}$$

This shows that $p_{X|Y}(x|12)$ is a hypergeometric probability mass function with parameters $D = 10$, $N = 100$, and $n = 12$ (see Definition 5.6 of the book). Hence its expected value is $nD/N = (12 \times 10)/100 = 1.2$; that is, $E(X | Y = 12) = 1.2$, an answer consistent with our intuitive expectation. ♦

Example 8f Let X and Y be two continuous random variables with the joint probability density function $f(x, y)$. Prove that for any subset A of real numbers,

- (i) $E(X | Y \in A) = \frac{1}{P(Y \in A)} \int_{-\infty}^{\infty} \left(\int_A x f(x, y) dy \right) dx.$
- (ii) $E(X | Y \in A) = \frac{1}{P(Y \in A)} \int_A E(X | Y = y) f_Y(y) dy.$

Proof: (i) Let $H(x) = P(X \leq x | Y \in A)$ and $h(x) = H'(x)$. Then

$$H(x) = \frac{P(X \leq x, Y \in A)}{P(Y \in A)} = \frac{1}{P(Y \in A)} \int_{-\infty}^x \left(\int_A f(x, y) dy \right) dx.$$

Now using fundamental theorem of calculus, we differentiate H to obtain h . We get

$$h(x) = \frac{1}{P(Y \in A)} \int_A f(x, y) dy.$$

Therefore,

$$E(X | Y \in A) = \int_{-\infty}^{\infty} xh(x) dx = \frac{1}{P(Y \in A)} \int_{-\infty}^{\infty} \left(\int_A x f(x, y) dy \right) dx.$$

(ii) By changing the order of integration in (i) we obtain

$$\begin{aligned} E(X | Y \in A) &= \frac{1}{P(Y \in A)} \int_A \left(\int_{-\infty}^{\infty} \frac{x f(x, y)}{f_Y(y)} dx \right) f_Y(y) dy \\ &= \frac{1}{P(Y \in A)} \int_A E(X | Y = y) f_Y(y) dy. \end{aligned}$$

Remark 8a Let a and b be two real numbers and $F(x, y)$ be the joint probability distribution function of X and Y . This theorem implies relations such as

$$E(X | a < Y < b) = \frac{1}{F_Y(b) - F_Y(a)} \int_{-\infty}^{\infty} \left(\int_a^b x f(x, y) dy \right) dx.$$

and

$$E(X | a < Y < b) = \frac{1}{F_Y(b) - F_Y(a)} \int_a^b E(X | Y = y) f_Y(y) dy. \quad \blacklozenge$$

Example 8g Let the joint probability density function of the continuous random variables X and Y be given by

$$f(x, y) = \begin{cases} \frac{20}{13}(x + y) & 0 \leq x \leq \sqrt{y} \leq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

- (i) Calculate $P(0 < X < 3/4, 1/4 < Y < 3/4)$.
- (ii) Find the marginal probability density functions of X and Y , the conditional probability density function of X given that $Y = y$, and the conditional probability density function of Y given that $X = x$.
- (iii) Calculate $E(X | Y = y)$ and $E(Y | X = x)$.

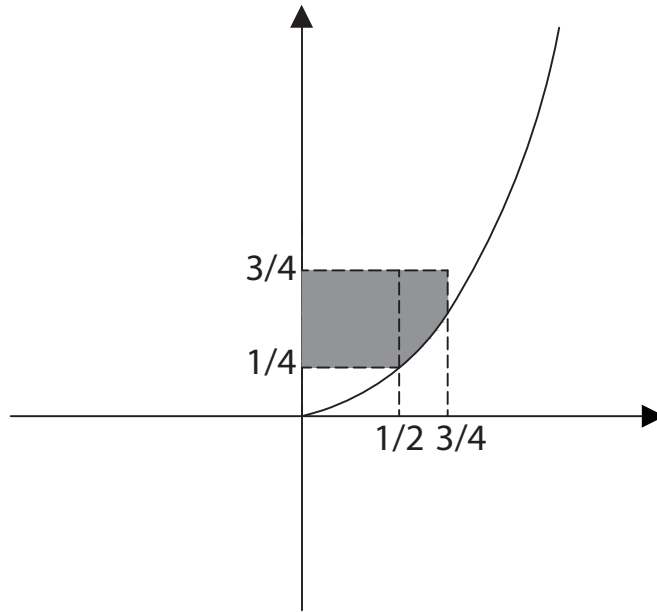


Figure 8g The shaded area is the region of integration of Example 8g.

Solution: (i) The region of integration for calculation of

$$P(0 < X < 3/4, 1/4 < Y < 3/4)$$

is shown in Figure 8g. Thus

$$\begin{aligned} P(0 < X < 3/4, 1/4 < Y < 3/4) &= \\ \int_0^{1/2} \left[\int_{1/4}^{3/4} \frac{20}{13}(x+y) dy \right] dx &+ \int_{1/2}^{3/4} \left[\int_{x^2}^{3/4} \frac{20}{13}(x+y) dy \right] dx = \\ \frac{5}{13} \int_0^{1/2} (2x+1) dx &+ \frac{5}{104} \int_{1/2}^{3/4} (-16x^4 - 32x^3 + 24x + 9) dx = \\ \frac{15}{52} + \frac{1059}{6656} &= \frac{2979}{6656} \approx 0.45. \end{aligned}$$

(ii) We have

$$f_X(x) = \int_{x^2}^1 \frac{20}{13}(x+y) dy = \frac{10}{13}(-x^4 - 2x^3 + 2x + 1), \quad 0 \leq x \leq 1;$$

$$f_Y(y) = \int_0^{\sqrt{y}} \frac{20}{13}(x+y) dx = \frac{10}{13}y(2\sqrt{y} + 1), \quad 0 \leq y \leq 1;$$

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{2(x+y)}{y(2\sqrt{y}+1)}, \quad 0 \leq x \leq \sqrt{y}, \quad 0 \leq y \leq 1;$$

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{2(x+y)}{-x^4 - 2x^3 + 2x + 1}, \quad x^2 \leq y \leq 1, \quad 0 \leq x \leq 1.$$

(iii) By relation (8.21) of the book,

$$\begin{aligned} E(X | Y = y) &= \int_0^{\sqrt{y}} x f_{X|Y}(x|y) dx \\ &= \int_0^{\sqrt{y}} \frac{2x(x+1)}{y(2\sqrt{y}+1)} dx = \frac{3y + 2\sqrt{y}}{3(2\sqrt{y}+1)}, \quad 0 \leq y \leq 1; \\ E(Y | X = x) &= \int_{x^2}^1 y f_{Y|X}(y|x) dy = \int_{x^2}^1 \frac{2y(x+y)}{-x^4 - 2x^3 + 2x + 1} dy \\ &= \frac{-2x^6 - 3x^5 + 3x + 2}{3(-x^4 - 2x^3 + 2x + 1)}, \quad 0 \leq x \leq 1. \quad \blacklozenge \end{aligned}$$

In Section 8.4 of the book, we used Theorem 8.8 to find the joint probability density function of functions of continuous random variables. Calculation of joint probability mass functions of functions of discrete random variables is done similarly. An example follows.

Example 8h Let X and Y be independent Poisson random variables with parameters λ_1 and λ_2 , respectively. Find the joint probability mass function of $X + Y$ and Y , and the probability mass function of $X + Y$.

Solution: Let $p(x, y)$ be the joint probability mass function of X and Y . Then

$$p(x, y) = \frac{\lambda_1^x e^{-\lambda_1}}{x!} \cdot \frac{\lambda_2^y e^{-\lambda_2}}{y!} = \frac{\lambda_1^x \lambda_2^y}{x! y!} e^{-(\lambda_1 + \lambda_2)},$$

$x = 0, 1, 2, \dots$, $y = 0, 1, 2, \dots$. To find $g(u, v)$, the joint probability mass function of $X + Y$ and Y , let $U = X + Y$ and $V = Y$, then $X = U - V$ and $Y = V$. So

$$g(u, v) = \frac{\lambda_1^{u-v} \lambda_2^v}{(u-v)! v!} e^{-(\lambda_1 + \lambda_2)},$$

$u = 0, 1, 2, \dots$, $v = 0, 1, 2, \dots$.

Note that $g_U(u)$, the marginal probability mass function of $g(u, v)$ gives the probability mass function of $X + Y$, the sum of two independent Poisson random variables. Since

$V < U$,

$$\begin{aligned} g_U(u) &= \sum_{v=0}^u g(u, v) = e^{-(\lambda_1 + \lambda_2)} \sum_{v=0}^u \frac{\lambda_1^{u-v} \lambda_2^v}{(u-v)! v!} \\ &= \frac{e^{-(\lambda_1 + \lambda_2)}}{u!} \sum_{v=0}^u \frac{u!}{(u-v)! v!} \lambda_1^{u-v} \lambda_2^v \\ &= \frac{e^{-(\lambda_1 + \lambda_2)}}{u!} (\lambda_1 + \lambda_2)^u. \end{aligned}$$

This shows that the sum of two independent Poisson random variables with parameters λ_1 and λ_2 is Poisson with parameters $\lambda_1 + \lambda_2$. We will prove this important fact again using a different method in Section 11.2 of the book (see Theorem 11.5). ♦

Example 8i Let X and Y be two independent beta random variables both with parameters $(2, 2)$. Find the joint probability density function of $U = X/Y$ and $V = XY$.

Solution: Let $f(x, y)$ be the joint probability density function of X and Y . We are given that

$$f_X(x) = \begin{cases} 6x(1-x) & 0 < x < 1 \\ 0 & \text{otherwise,} \end{cases}$$

and

$$f_Y(y) = \begin{cases} 6y(1-y) & 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Thus

$$f(x, y) = f_X(x)f_Y(y) = \begin{cases} 36xy(1-x)(1-y) & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Consider the system of two equations in two unknowns

$$\begin{cases} x/y = u \\ xy = v. \end{cases}$$

Since $x > 0$ and $y > 0$, this system has the unique solution

$$\begin{cases} x = \sqrt{uv} \\ y = \sqrt{v/u}. \end{cases}$$

Now $0 < x < 1$ implies that $v < 1/u$, and $0 < y < 1$ implies that $v < u$. So this system transforms $R = \{(x, y) : 0 < x < 1, 0 < y < 1\}$ to the shaded region of Figure 8h, which is

$$Q = \{(u, v) : 0 < u \leq 1, 0 < v < u\} \cup \{(u, v) : 1 < u < \infty, 0 < v < 1/u\}.$$

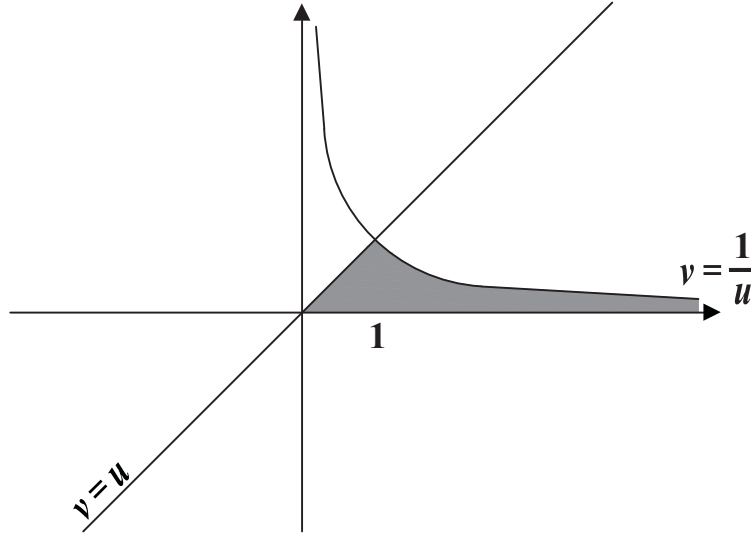


Figure 8h The shaded region is Q in Example 8i.

Since

$$J = \begin{vmatrix} \frac{1}{2}\sqrt{v/u} & \frac{1}{2}\sqrt{u/v} \\ -\frac{1}{2}\sqrt{v/u^3} & \frac{1}{2}\sqrt{1/uv} \end{vmatrix} = \frac{1}{2u} \neq 0,$$

by Theorem 8.8 of the book, $g(u, v)$, the joint probability density function of U and V , is given by

$$g(u, v) = f(u, v)|J| = \begin{cases} \frac{18v}{u}(1 - \sqrt{uv})\left(1 - \sqrt{\frac{v}{u}}\right) & \text{if } (u, v) \in Q \\ 0 & \text{elsewhere.} \end{cases} \quad \blacklozenge$$

Example 8j From a certain airport, starting midnight, Airline A has flights every 4 hours every day. From the same airport, starting Y hours after midnight, Airline B has flights every 6 hours every day. Suppose that Y is a random number from the interval $(0, 4)$; that is, the first flight of the Airline B is at a random time between midnight and 4:00 A. M. Find the probability that a passenger who arrives at this airport at a random time finds that the next departure is from Airline A .

Solution: On a given day, the number of hours after midnight that Airline A has departures are 0, 4, 8, 12, 16, and 20, and the number of hours after midnight that Airline B has departures are $Y, Y + 6, Y + 12$, and $Y + 18$, where Y is a uniform random variable over the interval $(0, 4)$. By looking at these departure times carefully, we observe that, after every 12 hours, the system restarts probabilistically all over again. That is, if the 12th hour

is taken as 0, then again the departure times of Airline *A* will be 0, 4, 8, 12, 16, and 20, and the departure times of Airline *B* will be Y , $Y + 6$, $Y + 12$, and $Y + 18$, which is a replica of the original situation. By this observation, the problem can be rephrased as follows:

*In the interval $[0, 12]$, Airline *A* has flights at 0, 4, 8, and 12. Airline *B* has flights at Y and $Y + 6$, where Y is a random number from the interval $(0, 4)$. If the time interval $[0, 12]$ is observed at a random time X , what is the probability that the first departure after X is from Airline *A*?*

To answer this, note that, if $0 < Y \leq 2$, then in the period $[0, 12]$, the departure times for flights from Airlines *A* and *B*, combined, are 0, Y , 4, $Y + 6$, 8, and 12. Hence the first departure after X is from *A* if and only if

$$Y \leq X < 4 \quad \text{or} \quad Y + 6 \leq X < 12.$$

If $Y > 2$, then, in $[0, 12]$, the departure times of flights *A* and *B*, combined, are 0, Y , 4, 8, $Y + 6$, and 12. Therefore, in this case, the first departure after X is from *A* if and only if

$$Y \leq X < 8 \quad \text{or} \quad Y + 6 \leq X < 12.$$

Let

$$R_1 = \{(x, y) : 0 < y < 2\} \cap \{(x, y) : y \leq x < 4 \text{ or } y + 6 \leq x < 12\},$$

and

$$R_2 = \{(x, y) : y > 2\} \cap \{(x, y) : y \leq x < 8 \text{ or } y + 6 \leq x < 12\}.$$

Then the set $R_1 \cup R_2$ is the shaded region of Figure 8i.

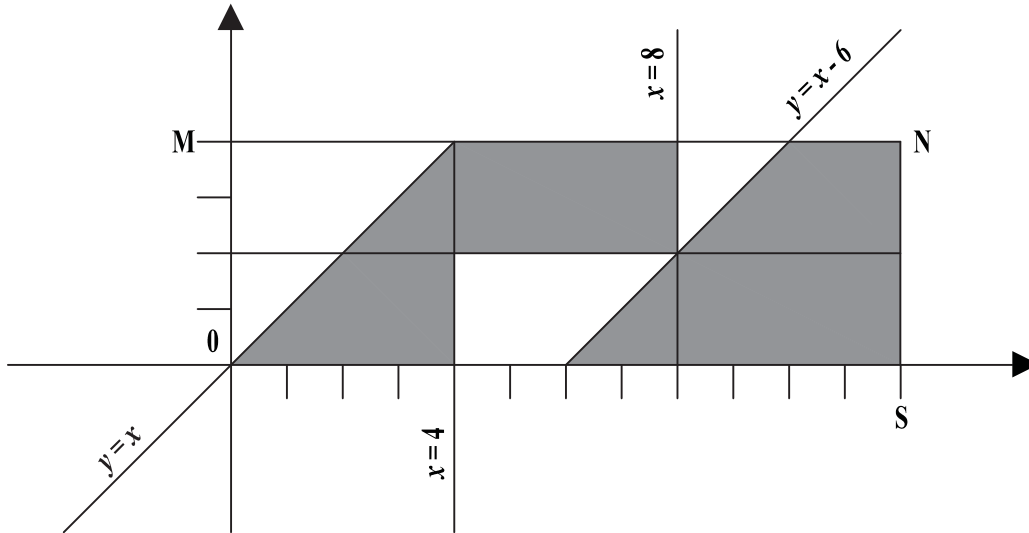


Figure 8i Geometric Model of Example 8j.

Clearly, the desired probability is the area of $R_1 \cup R_2$ divided by 48, the area of the rectangle OMNS, which is the sample space. But the area of $R_1 \cup R_2$ is the summation of the areas of the three shaded trapezoids, which is easily calculated and found to be 32. Hence the desired probability is $32/48=2/3$. ♦

Example 8k Let X and Y be two independent random variables with the same probability density function, given by

$$f(x) = \begin{cases} 1/x^2 & \text{if } x \geq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

- (i) Calculate the probability density function of X/Y directly, without using Theorem 8.8 of the book.
- (ii) Find $P(X < 3Y)$.

Solution: Since X and Y are independent and have the same probability density function,

$$f(x, y) = f_X(x)f_Y(y),$$

where,

$$f_X(x) = \begin{cases} 1/x^2 & \text{if } x \geq 1 \\ 0 & \text{elsewhere,} \end{cases}$$

and

$$f_Y(y) = \begin{cases} 1/y^2 & \text{if } y \geq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Thus

$$f(x, y) = f_X(x)f_Y(y) = \begin{cases} 1/(x^2y^2) & \text{if } x \geq 1, y \geq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Let G be the distribution function of X/Y . We will first calculate G and then, by differentiation, find g , the probability density function of X/Y . Clearly, for $t \leq 0$, $G(t) = 0$. If $0 < t < 1$, then the event $X/Y \leq t$ occurs if and only if $X \leq tY$. Since $X \geq 1$ implies that $Y \geq 1/t$, we have that $X \leq tY$ is equivalent to $Y \geq 1/t$ and $1 \leq X \leq tY$. As Figure 8j shows, for $0 < t < 1$,

$$\begin{aligned} G(t) &= P(X/Y \leq t) = P(X \leq tY) = P\left(Y \geq \frac{1}{t} \text{ and } 1 \leq X \leq tY\right) \\ &= P\left((X, Y) \in \left\{(x, y): 1 \leq x \leq ty, y \geq \frac{1}{t}\right\}\right) \\ &= \int_{1/t}^{\infty} \left(\int_1^{ty} \frac{1}{x^2y^2} dx \right) dy = \int_{1/t}^{\infty} \left[-\frac{1}{xy^2} \right]_1^{ty} dy \end{aligned}$$

$$= \int_{1/t}^{\infty} \left(-\frac{1}{ty^3} + \frac{1}{y^2} \right) dy = \left[\frac{1}{2ty^2} - \frac{1}{y} \right]_{1/t}^{\infty} = t/2.$$

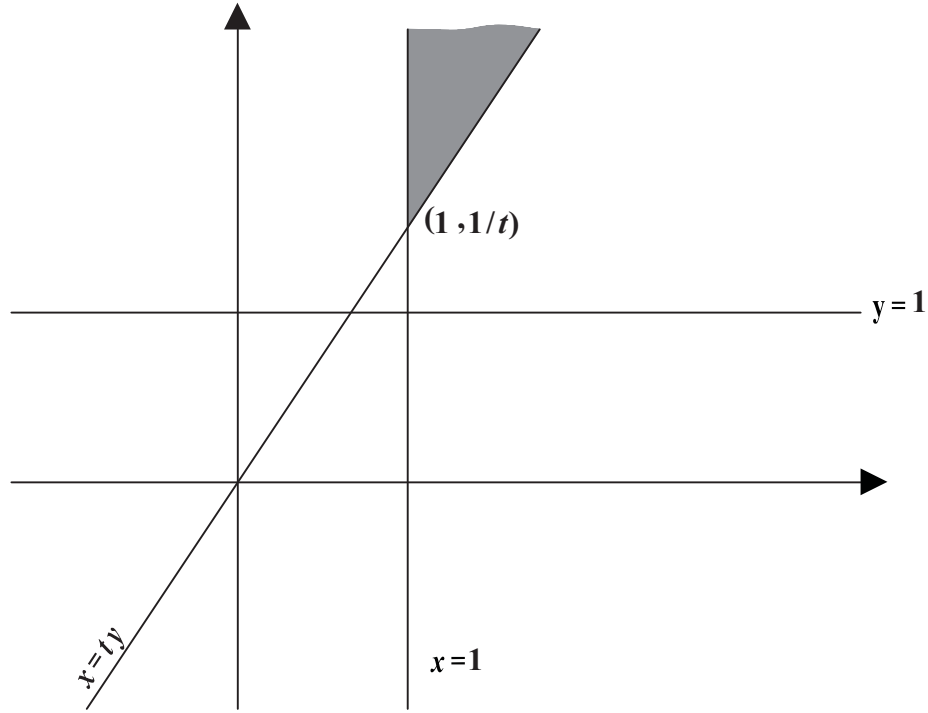


Figure 8j The shaded area is the region of integration of Example 8k, $0 < t < 1$.

As Figure 8k shows, for $t \geq 1$,

$$\begin{aligned} G(t) &= P(X \leq tY) = P\left((X, Y) \in \{(x, y) : 1 < x \leq ty, y > 1\}\right) \\ &= \int_1^{\infty} \left(\int_1^{ty} \frac{1}{x^2 y^2} dx \right) dy = \int_1^{\infty} \left(-\frac{1}{ty^3} + \frac{1}{y^2} \right) dy \\ &= \left[\frac{1}{2ty^2} - \frac{1}{y} \right]_1^{\infty} = 1 - \frac{1}{2t}. \end{aligned}$$

Thus,

$$G(t) = \begin{cases} 0 & t < 0 \\ \frac{t}{2} & 0 \leq t < 1 \\ 1 - \frac{1}{2t} & t \geq 1, \end{cases}$$

and $g(t) = G'(t)$ is given by

$$g(t) = \begin{cases} 0 & t < 0 \\ \frac{1}{2} & 0 \leq t < 1 \\ \frac{1}{2t^2} & t \geq 1. \end{cases}$$

(ii) $P(X < 3Y) = P(X/Y < 3) = G(3) = 1 - \frac{1}{6} = \frac{5}{6}. \quad \blacklozenge$

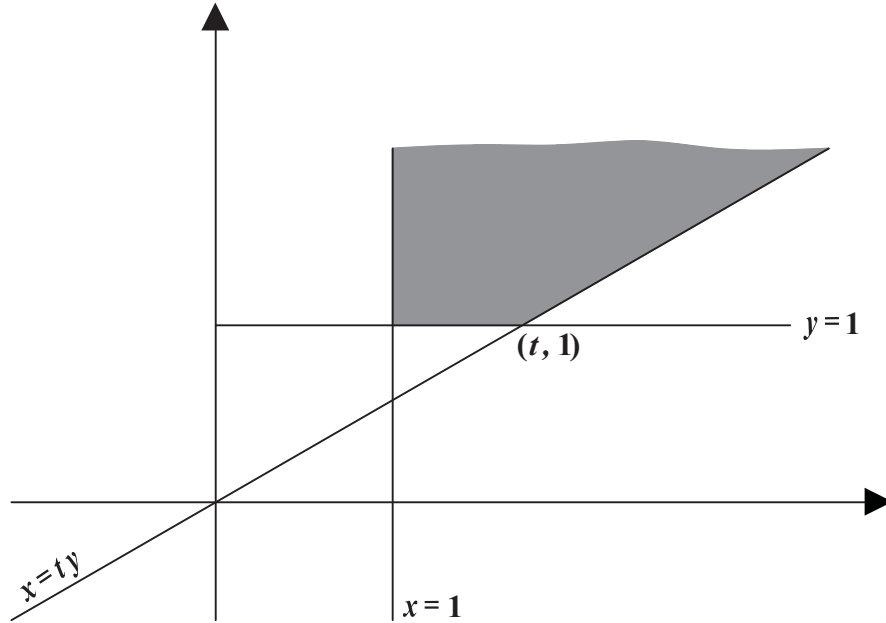


Figure 8k The shaded area is the region of integration of Example 8k, $t \geq 1$.

Example 8ℓ For an integer $n \geq 3$, let X be a standard normal random variable and Y be a gamma random variable with parameters $(n/2, 1/2)$. Suppose that X and Y are independent and $U = X/\sqrt{Y/n}$. The random variable U , called t -distributed with n degrees of freedom, has significant applications in statistics. Find the probability density function of U .

Solution: Let $f(x, y)$ be the joint probability density function of X and Y . Since X and Y are independent, $f(x, y) = f_X(x)f_Y(y)$, where

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty$$

$$f_Y(y) = \frac{y^{(n/2)-1} e^{-y/2}}{2^{n/2} \Gamma(n/2)}, \quad y > 0.$$

Thus

$$f(x, y) = \frac{1}{\sqrt{2\pi} \cdot 2^{n/2} \cdot \Gamma(n/2)} \cdot y^{(n/2)-1} \cdot e^{-(x^2+y)/2}, \quad -\infty < x < \infty, y > 0.$$

Let $V = Y$. We will find the joint probability density function of U and V . The desired function is the marginal probability density function of U . The system of two equations in two unknowns

$$\begin{cases} \frac{x}{\sqrt{y/n}} = u \\ y = v \end{cases}$$

defines a one-to-one transformation of $R = \{(x, y): -\infty < x < \infty, y > 0\}$ onto the region $Q = \{(u, v): -\infty < u < \infty, v > 0\}$. It has the unique solution $x = u\sqrt{v/n}$, $y = v$. Hence

$$\mathbf{J} = \begin{vmatrix} \sqrt{v/n} & \frac{u}{2\sqrt{nv}} \\ 0 & 1 \end{vmatrix} = \sqrt{v/n} \neq 0.$$

By Theorem 8.8 of the book, $g(u, v)$, the joint probability density function of U and V is given by

$$\begin{aligned} g(u, v) &= f(u\sqrt{v/n}, v) \sqrt{v/n} \\ &= \frac{1}{\sqrt{2\pi n} \Gamma(n/2) 2^{n/2}} v^{[(n+1)/2]-1} \exp \left[-\frac{v}{2} \left(1 + \frac{u^2}{n} \right) \right], \quad -\infty < u < \infty, v > 0. \end{aligned}$$

For simplicity, let $k = \frac{1}{\sqrt{2\pi n} \Gamma(n/2) 2^{n/2}}$. Then the probability density function of U is $g_U(u)$, where

$$g_U(u) = k \int_0^\infty v^{[(n+1)/2]-1} \exp \left[-\frac{v}{2} \left(1 + \frac{u^2}{n} \right) \right] dv.$$

Let $\frac{v}{2} \left(1 + \frac{u^2}{n} \right) = w$. Then $v = \frac{2}{1 + (u^2/n)} w$ and $dv = \frac{2}{1 + (u^2/n)} dw$. Thus

$$\begin{aligned} g_U(u) &= k \int_0^\infty \frac{2^{[(n+1)/2]-1}}{[1 + (u^2/n)]^{[(n+1)/2]-1}} w^{[(n+1)/2]-1} e^{-w} \cdot \frac{2}{1 + (u^2/n)} dw \\ &= k \cdot \frac{2^{(n+1)/2}}{[1 + (u^2/n)]^{(n+1)/2}} \int_0^\infty w^{[(n+1)/2]-1} e^{-w} dw \\ &= k \cdot \frac{2^{(n+1)/2}}{[1 + (u^2/n)]^{(n+1)/2}} \Gamma\left(\frac{n+1}{2}\right) \end{aligned}$$

$$= \frac{1}{\sqrt{2\pi n} \Gamma(n/2) 2^{n/2}} \cdot \frac{2^{(n+1)/2}}{[1 + (u^2/n)]^{(n+1)/2}} \Gamma\left(\frac{n+1}{2}\right).$$

Simplifying the last term, we have that the probability density function of U is given by

$$g_U(u) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{\pi n} \Gamma(n/2)} \cdot \frac{1}{[1 + (u^2/n)]^{(n+1)/2}}, \quad -\infty < u < \infty. \quad \blacklozenge$$